

Resume — Inhyeok Hwang

INHYEOK HWANG

Senior Researcher · Electrical Control & Product Development (Junior PM, concurrent)

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SUMMARY

Reliability-focused mechanical engineer — M.S. from Konkuk University's Reliability-Based Design Optimization (RBDO) Lab, where reliability was the thesis topic, not a side topic. Thesis research on IPMSM digital-twin fault diagnosis (Co-simulation of Ansys Maxwell FEM and a MATLAB/Simulink inverter model). Three peer-reviewed journal publications as co-author (IEEE Transactions on Instrumentation and Measurement · Solar Energy · Journal of Power Electronics, all 2024) and two Korean patents.

Three and a half years of industrial product development at GINT Corp., leading reliability testing (DVT/PVT-equivalent within APQP Phase 2–3) for the GT-SS500 48 V autonomous speed sprayer, with Phase 1/4/5 supporting scope across the full program lifecycle. Authored DFMEA #201/#210 Step 1–7 with five high-priority risks closed before ramp, applied ISO 25119 AgPL d assessment to safety functions, led deep failure analysis on four real field issues (Fishbone-driven root-cause analysis), and tracked 27 NCRs through closure. Built four custom test rigs in-house (dynamometer, fan bench, pump bench, bumper safety rig) and coordinated a structured 48 V BMS supplier quality review.

Independently built a physics-based simulation and test-automation stack for the same vehicle class — 497 automated tests under CI — to exercise field-usage scenarios and protection-logic failure modes ahead of physical testing.

EXPERIENCE

GINT Corp. — Electrical Control R&D + Product Development (Junior PM, concurrent)

Senior Researcher · Jan 2025 – Present

GT-SS500: 48 V Electric Autonomous Speed Sprayer — Full Lifecycle

- Led **APQP Phase 2–3** (primary scope: design validation, DFMEA, IQC/OQC criteria, NCR closed-loop) for GT-SS500, coordinating six cross-functional teams from algorithm development to a first-article production plan of **16 units + 2 transport carts + 2 spares**
- Supporting scope across the full program lifecycle: **Phase 1** (MTBF/B10 reliability goals, VOC, special-characteristics identification), **Phase 4** (PPAP ROM sign-off), **Phase 5** (37+ field-issue database; cascading-failure analysis identified six failure chains across alpha-prototype issues)
- Authored **DFMEA #201 and #210 (Step 1–7)** and applied **ISO 25119 AgPL** risk assessment — over-charge protection rated **AgPL d** (highest category for agricultural machinery functional safety); five high-priority (AP=H) risks resolved before production ramp
- Led **deep failure analysis on four real field issues**, applying Fishbone (Ishikawa) cause-and-effect analysis to isolate root cause and mapping corrective actions back into DFMEA:
- MCB contact carbonization (#204)**: confirmed electrolytic corrosion mechanism; breaker replaced with IEC 60947-2-grade part and re-qualified
- LCD-mode state-machine bug (#79)**: undefined transitions removed; verified through validation runs
- Pump O-ring freeze failure**: O-ring redesigned and re-qualified for low-temperature operation
- Ground-bounce noise on the control bus**: countermeasure mapped back into the DFMEA
- Operated **27 NCRs** through to closure as a closed-loop corrective action process (Failure → Reporting → Analysis → Corrective Action → System update); defined IQC/OQC acceptance criteria with SPC-based **Cpk ≥ 1.33** (short-term capability target) for key characteristics
- Managed a structured quality review when our 48 V BMS supplier visited our facility — 6-item agenda covering FW version verification, SOC anomaly root-cause, and relay-behavior confirmation
- Currently mentoring a **junior engineer** in embedded control-system design — CAN protocol architecture, state-machine safety design, and DFMEA methodology
- Built **four custom test rigs** in-house: **dynamometer torque control (nonlinearity 0.008%)**, **fan bench (airflow 7.9 → 12.4 m/s, +57%)**, **pump bench** (three-sample linearity characterization, nameplate-spec falsification), **bumper safety**

rig (stop distance 0.082 m, peak current 308 A, settling time 467 ms at 3 km/h)

- Designed a **5-node CAN distributed-control architecture** (VCU ↔ dual drive MC ↔ fan ESC ↔ pump DRV ↔ BMS); authored four DBC files; detected four breaking changes before integration through a three-pass consistency protocol
- **BOM management** across 132 items; 23 items with insufficient stock surfaced ahead of ramp
- Personally resolved **14 field issues** in the electrical-control scope (subset of the Phase 5 database above); represented program at Innovation-Product on-site audit (4 of 9 items handled)

GINT Corp. — Electrical Control R&D

Junior Researcher · Feb 2023 – Dec 2024

EOP 400 W: Automotive BLDC Motor Control (Government R&D Project, NXP MC9S12ZVMC)

- Co-authored a programmable online bond-wire fault detection and location method for IGBT power semiconductor modules using inverter output parameters — published in *IEEE Transactions on Instrumentation and Measurement* (2024, third author, DOI 10.1109/TIM.2024.3472910)
- **SVPWM/DPWM switching strategy**: FET temperature reduced 1–6 °C, input power reduced 1–3.8 % (validated across 2,932 data points)
- **Cryogenic start test**: cold-start time reduced from 300 s to 100 s at –40 °C
- **CAN Sleep/Wakeup**: quiescent-current-compliant bus management (five revision cycles)
- Built a **dynamometer torque-control system** with nonlinearity 0.008 %; presented at internal seminar
- Two Korean patent contributions (see Patents)

SELECTED INDEPENDENT PROJECT

RS500 — Physics Simulation & Test-Automation Stack for a Tracked Autonomous Vehicle

Feb 2026 – Aug 2026 · open source: github.com/hwanginhyeok/physical_AI_rs500 · Python / ROS 2 / Gazebo

Built, outside work, a Level-2 physics simulation of the SS500 vehicle class (dual 3 kW BLDC, dual-channel motor driver, 48 V LiFePO4) so that usage scenarios and protection-logic failure modes could be exercised before hardware time.

- **Vehicle & drivetrain model**: skid-steer kinematics (ICR), Newtonian dynamics with load transfer and grade, motor + reducer first-order lag with driver acceleration/deceleration limits, **motor-driver protection modeled as failure modes (stall, over-current, over-temperature, over-voltage)**, CAN codec for the driver protocol
- **Environment & sensing**: Bekker terramechanics for track–soil interaction, GPS/IMU noise model, dual-EKF sensor fusion, LiDAR obstacle detection and camera object detection
- **Usage-model-driven batch testing**: scenario runner sweeping **3 mission profiles × 5 terrain types**, and a module-level **parameter-sensitivity report generator** for drivetrain, terrain, dynamics and sensor-noise blocks — the same stress-factor screening logic used to size accelerated tests
- **Verification**: **497 automated tests** across six packages (core physics 331, perception 72, planning 44, control 26, CAN bridge 24), run under GitHub Actions CI on every commit; ROS 2 bring-up with Gazebo, Nav2 and Foxglove

EDUCATION

Konkuk University — M.S. in Mechanical Design (RBDO Lab)

Mar 2021 – Feb 2023 · 27 major credits · Major GPA 3.77/4.5 · Advisor: Prof. Namsu Kim

Thesis: *Fault Diagnosis Simulation of Interior Permanent Magnet Synchronous Motor using Electromagnetic Analysis* (49 pp., RISS f678963f23f2e418ffe0bdc3ef48d419)

- Built an IPMSM digital twin: flux-state variable model + Ansys Maxwell FEM, co-simulated with a MATLAB/Simulink inverter model and validated against experimental data across multiple operating conditions
- Participated in a government-funded PHM SoC development project (MOTIE, 2021–2022); contributed to multi-sensor fault diagnosis and remaining-useful-life estimation work for the IPMSM drivetrain
- Observed an IPMSM accelerated-life-test program (eccentricity-fault characterization with load-factor stress); the resulting work was later published as *Identification of failure modes in IPMSM under accelerated life test based on dual sensor architecture* (J. Power Electronics, 2024, sixth author)

Konkuk University — B.S. in Mechanical Design

Mar 2015 – Feb 2021 · 79 major credits · Major GPA 3.24/4.5

PUBLICATIONS & PATENTS

Peer-Reviewed Journals (3, co-authored)

- **IEEE Trans. on Instrumentation and Measurement** 73 (2024), Art. 10726721 — *Programmable Online Bond-Wire Fault Detection and Location Method for IGBT Using Inverter Output Parameters* · DOI 10.1109/TIM.2024.3472910 · **3rd author**
- **Solar Energy** 276 (2024), Art. 112645 — *Lifetime prediction of polymeric materials in PV module under continuously varying environments based on damage summation approach* · DOI 10.1016/j.solener.2024.112645 · 4th author
- **Journal of Power Electronics** 24(5) (2024), 822–831 — *Identification of failure modes in IPMSM under accelerated life test based on dual sensor architecture* · DOI 10.1007/s43236-024-00810-8 · 6th author

Conference Presentations (4)

- 2023 · PHM Society Asia-Pacific Conference — 120 kW IPMSM system-level fault diagnosis (DOI 10.36001/phmap.2023.v4i1.3780)
- 2022 · PCIM Asia — IGBT IPM power cycling: bond-wire / solder degradation
- 2022 · Korean Society for Reliability, Spring Conference — Predictive maintenance for EV drivetrain · **Best Paper Award**
- 2021 · Korean PHM Society Annual Conference — IGBT open-fault diagnosis via system-level measurements · **Best Poster Award**

Patents (2)

- **Patent #1** — Motor initial position detection (co-inventor) · KR App. PN231067KR (2023, filed via The Wave IP) · detection time 1.0 s → **0.56 s** · prior-art screen of 1 paper and 10 patents
- **Patent #2** — *Method for preventing step-out of rotor of motor in low temperature and apparatus thereof* · KR App. **10-2023-0175484** (filed 2023-12-06) · Reg. **10-2654562** (granted 2024-04-01) · Applicant: GINT Inc. · IPC H02P 29/032 · **Development contributor** (experimental validation of -40 °C cold start, 300 s → 100 s; not on the inventor list — inventors of record: Je Jeong-mun, Kim Eun-tae)

SKILLS

Motor Control & Power Electronics

- BLDC · IPMSM · PMSM control · FOC · SVPWM/DPWM switching · sensorless initial-position estimation · BEMF observer · cryogenic start
- Inverter gate-signal diagnosis · IGBT bond-wire lift-off detection (IEEE TIM 2024)

Embedded & Firmware

- STM32 · NXP MC9S12ZVMC · Infineon TC23x · RTOS · multi-mode state machines (RC / LCD / autonomous) · motor-controller firmware code review

Communication & System Design

- CAN · CAN-FD · CAN Sleep/Wakeup · DBC authoring (4 files) · CAN BusLoad analysis · 5-node distributed control topology
- Reading-level familiarity with ISO 13849 and functional-safety mapping

Simulation & Analysis

- Ansys Maxwell (FEM, transient) · MATLAB/Simulink · Co-simulation (Sim2Real precursor) · MIL/SIL/HIL concepts
- Vehicle-level physics simulation in Python (dynamics, terramechanics, sensor-noise, protection-logic failure modes) · ROS 2 / Gazebo / Nav2 · batch scenario and parameter-sensitivity tooling (RS500 project)

Test & Measurement

- Dynamometer torque control · self-built fan / pump / bumper benches · CANoe · Vector VN1600 · SPICE · oscilloscope · NI DAQ (NI-9215)
- Test automation: pytest suites under CI (497 tests, RS500) · scenario sweeps as usage-model stress screening

PHM & Signal Processing (Machine Learning for Reliability)

- **Government-funded PHM SoC development** (MOTIE, 2021–2022): the team built an SVM fault classifier on phase-current signatures for demagnetization & winding-fault diagnosis of a 120 kW IPMSM. **My contribution:** the physics-based co-simulation (Ansys Maxwell FEM + MATLAB/Simulink) that generated the labeled training data, and the motor / fault modeling underneath it
- Multi-sensor fusion (phase current · temperature · shaft displacement · vibration) · anomaly detection · familiarity with field product usage model → ALT stress-factor mapping (co-authored Solar Energy 2024 on continuously varying-

environment damage summation)

Quality & PM Artifacts

- APQP Phase 2–3 (primary) + Phase 1/4/5 supporting · DFMEA / PFMEA · DRBFM · boundary diagram · BOM management · IQC/OQC inspection guides with **Cpk $\geq$ 1.33 acceptance criteria** · WBS · NCR closed-loop handling (FRACAS-equivalent)

Software & Engineering Automation

- Python · C/C++ · Git · Linux · pytest / GitHub Actions CI
- Built and operate a personal engineering-automation fleet with the same verification discipline as test work: 86 scheduled jobs and 37 service units under drift monitoring; a task runtime that verifies delivered artifacts against declared scope (3,058 tasks, Aug–Sep 2026); a document index with measured retrieval quality (MRR 0.44, 26-question golden set)
- Learning / applying: PyTorch · scikit-learn · Docker

Domain Study

- Maintain a structured engineering knowledge base — 119 notes across control, electrical, mechanical, automotive quality, robotics and physics (concept → equation → applied case), 46 of them cross-checked against primary sources

Languages

- Korean: native
- English: professional working proficiency
- TOEIC 920 / TOEIC Speaking Lv. 6 (IH 140) (expired)

AWARDS

- **Best Paper Award**, Korean Society for Reliability — Spring Conference 2022 (predictive maintenance for EV drivetrain)
- **Best Poster Award**, Korean PHM Society — Annual Conference 2021 (IGBT open-fault diagnosis via system-level measurements)